

LISTA 8

1-a) $X = (-5, \frac{2}{3}, 0) + \lambda(\frac{1}{2}, 1, 1)$; $S: z = 3x - 2y - 16$
 $\vec{v} = (\frac{1}{2}, 1, 1)$ $z = 3x$ $y = \frac{3x+16}{2}$

$\vec{v}_r \times \vec{v}_s = \begin{vmatrix} i & j & k \\ \frac{1}{2} & 1 & 1 \\ 1 & \frac{3}{2} & 3 \end{vmatrix} = (\frac{3}{2}, -\frac{1}{2}, -\frac{1}{4})$ $X = (0, 8, 0) + \lambda(1, \frac{3}{2}, 3)$
 $\vec{v}_s = (1, \frac{3}{2}, 3)$

$\|\vec{v}_r \times \vec{v}_s\|^2 = (\frac{3}{2})^2 + (-\frac{1}{2})^2 + (-\frac{1}{4})^2$
 $\frac{9}{4} - \frac{1}{4} - \frac{1}{16} = \frac{141}{16} \Rightarrow \sqrt{\frac{141}{16}} = \frac{\sqrt{141}}{4}$

$\|\vec{v}_r\|^2 = (\frac{1}{2})^2 + 1^2 + 1^2 = \frac{9}{4} \Rightarrow \sqrt{\frac{9}{4}} = \frac{3}{2}$

$\|\vec{v}_s\|^2 = 1^2 + (\frac{3}{2})^2 + 3^2 = \frac{49}{4} \Rightarrow \sqrt{\frac{49}{4}} = \frac{7}{2}$

$\sin \alpha = \frac{\|\vec{v}_r \times \vec{v}_s\|}{\|\vec{v}_r\| \cdot \|\vec{v}_s\|} = \frac{\frac{\sqrt{141}}{4}}{\frac{3}{2} \cdot \frac{7}{2}} = \frac{\sqrt{141}}{9} = \frac{\sqrt{141}}{21}$

b) $r: X = (1, 1, 0) + \lambda(0, -1, 1)$; $S: x - y + 3 = z = 4$
 $\vec{v}_r = (0, -1, 1)$ $z = 4$; $y = x - 1$

$X = (1, 0, 4) + \lambda(1, 1, 0)$

$\vec{v}_s = (1, 1, 0)$

$n: \begin{vmatrix} i & j & k \\ 0 & -1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = (-1, 1, 1)$

$\|\vec{v}_r \times \vec{v}_s\| = \sqrt{(-1)^2 + 1^2 + 1^2} = \sqrt{3}$

$\|\vec{v}_r\| = \sqrt{0^2 + (-1)^2 + 1^2} = \sqrt{2}$

$\|\vec{v}_s\| = \sqrt{1^2 + 1^2 + 0} = \sqrt{2}$

$\sin \alpha = \frac{\|\vec{v}_r \times \vec{v}_s\|}{\|\vec{v}_r\| \cdot \|\vec{v}_s\|} = \frac{\sqrt{3}}{\sqrt{2} \sqrt{2}} = \frac{\sqrt{3}}{2}$

c) $r: \begin{cases} x + 3z = 7 \\ y = 0 \end{cases}$

$s: \begin{cases} x - 4y - 2z = 5 \\ y = 0 \end{cases}$

$x = 7 - 3z$

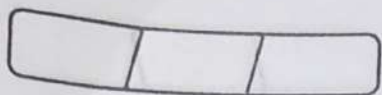
$x - 2z = 5 \Rightarrow x = 5 + 2z$

$r: X = (7, 0, 0) + \lambda(-3, 0, 1)$

$s: X = (5, 0, 0) + \lambda(2, 0, 1)$

$\vec{v}_r = (-3, 0, 1)$

$\vec{v}_s = (2, 0, 1)$



$$\vec{v}_r \cdot \vec{v}_s \Rightarrow (-3) \cdot 2 + 0 \cdot 0 + 1 \cdot 1 = -6 + 0 + 1 = -5$$

$$\|\vec{v}_r\| = \sqrt{(-3)^2 + 0^2 + 1^2} = \sqrt{10}$$

$$\|\vec{v}_s\| = \sqrt{2^2 + 0^2 + 1^2} = \sqrt{5}$$

$$\cos \alpha = \frac{\vec{v}_s \cdot \vec{v}_r}{\|\vec{v}_s\| \|\vec{v}_r\|} = \frac{-5}{\sqrt{10} \cdot \sqrt{5}} = \frac{-5}{\sqrt{50}} = \frac{-5}{5\sqrt{2}} = \frac{-1}{\sqrt{2}}$$

$$\Rightarrow \frac{-5}{5\sqrt{2}} = \frac{-1}{\sqrt{2}} = \frac{-\sqrt{2}}{2}$$

$$d) r = x = \frac{1-y}{2} = \frac{t}{3}$$

$$x=1, y=1+2\lambda, z=3\lambda$$

$$x = (0, 1, 0) + \lambda(1, -2, 3)$$

$$\vec{v}_r = (1, -2, 3)$$

$$s = \begin{cases} 3x + y - 5z = 0 \\ x - 2y + 3z + 1 = 0 \end{cases}$$

$$y = -3x + 5z \quad x = t - \frac{1}{4}$$

$$x = (-\frac{1}{4}, \frac{3}{4}, 0) + \mu(1, 2, 1)$$

$$\vec{v}_s = (1, 2, 1)$$

$$\vec{v}_r \cdot \vec{v}_s = 1 \cdot 1 + (-2) \cdot 2 + 3 \cdot 1 = 0$$

$$\cos \alpha = 0 = 90^\circ$$

$$2. r = x = (0, 2, 0) + \lambda(0, 1, 0) \text{ e ponto } q \quad s: x = (1, 2, 0) + \mu(0, 0, 1)$$

$$\vec{v}_r = (0, 1, 0)$$

$$\vec{v}_s = (0, 0, 1)$$

$$\text{Ponto genérico } (0, 2+\lambda, 0)$$

$$\text{Ponto genérico } (1, 2, \mu)$$

$$\vec{PQ} = Q - P = (1, -\lambda, \mu)$$

$$45^\circ: \cos 45^\circ = \frac{\vec{PQ} \cdot \vec{v}_r}{\|\vec{PQ}\| \cdot \|\vec{v}_r\|}$$

$$\frac{\sqrt{2}}{2} = \frac{(1, -\lambda, \mu) \cdot (0, 1, 0)}{\sqrt{1+\lambda^2+\mu^2} \cdot 1} = \frac{-\lambda}{\sqrt{1+\lambda^2+\mu^2}}$$

$$\frac{1}{2} = \frac{1+\lambda^2+\mu^2+2\lambda^2}{\sqrt{1+\lambda^2+\mu^2}} \Rightarrow \mu^2 = \lambda^2 - 1$$

$$\frac{1}{2} = \frac{1+\lambda^2+\mu^2+2\lambda^2}{\sqrt{1+\lambda^2+\mu^2}} \Rightarrow \mu^2 = \lambda^2 - 1$$

$$60^\circ: \cos 60^\circ = \frac{\vec{PQ} \cdot \vec{v}_s}{\|\vec{PQ}\| \cdot \|\vec{v}_s\|}$$

$$\frac{1}{2} = \frac{(1, -\lambda, \mu) \cdot (0, 0, 1)}{\sqrt{1+\lambda^2+\mu^2} \cdot 1} = \frac{\mu}{\sqrt{1+\lambda^2+\mu^2}}$$

$$\frac{1}{4} = 1 + \lambda^2 + \mu^2 = 4\mu^2 \Rightarrow \lambda^2 = 3\mu^2 - 1$$

$$\lambda^2 = 3(\lambda^2 - 1) - 1$$

$$-2\lambda^2 = -4$$

$$\lambda^2 = \pm \sqrt{2}$$

$$\text{para } \lambda = \sqrt{2}$$

$$M^2 = (\sqrt{2})^2 - 1 = 1 \Rightarrow M = \pm 1$$

$$\text{para } \lambda = -\sqrt{2}$$

$$M^2 = (-\sqrt{2})^2 - 1 = 1 \Rightarrow M = \pm 1$$

$$\text{Case } \lambda = \sqrt{2} \text{ e } M = 1$$

$$\cos \theta = \frac{(1, -\sqrt{2}, 1) \cdot (0, 1, 0)}{\sqrt{1+2+1}} = -\frac{\sqrt{2}}{2} \quad (\text{Não satisfaz})$$

$$\text{Case } \lambda = \sqrt{2} \text{ e } M = -1$$

$$\cos \theta = -\frac{\sqrt{2}}{2} \neq \cos 45^\circ \quad (\text{Não satisfaz})$$

$$\text{Case } \lambda = -\sqrt{2} \text{ e } M = 1$$

$$\cos \theta = \frac{(1, \sqrt{2}, 1) \cdot (0, 1, 0)}{\sqrt{1+2+1}} = \frac{\sqrt{2}}{2} \Rightarrow \theta = 45^\circ \text{ satisfaz}$$

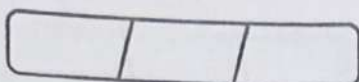
$$\cos \theta = \frac{1 + \sqrt{2} + 1}{2 \cdot 1} = \frac{1}{2} \Rightarrow \theta = 60^\circ$$

$$\text{Case } \lambda = -\sqrt{2} \text{ e } M = -1$$

$$\cos \theta = \frac{1 - \sqrt{2} - 1}{\sqrt{1+2+1}} = -\frac{\sqrt{2}}{2} \Rightarrow \theta = 45^\circ \text{ satisfaz}$$

$$\cos \theta = \frac{1 - \sqrt{2} - 1}{2 \cdot 1} = -\frac{1}{2} \neq \cos 60^\circ \quad \text{Não satisfaz}$$

$$S: P = (0, 2 - \sqrt{2}, 0) \text{ e } Q = (1, 2, 1)$$



3-a) $V: x = y = z = 0$

$$\pi = z = 0$$

$$V(0,0,0) \quad \vec{V} = (1,0,0) \quad (0,1,-1)$$

$$\pi = n\pi = (0,0,1)$$

$$\vec{V} = (1,0,0) \times (0,1,-1)$$

$$\begin{vmatrix} i & j & k \\ 1 & 0 & 0 \\ 0 & 1 & -1 \end{vmatrix} = (0,1,1)$$

$$\sin \theta = \frac{|(0,1,1) \cdot (0,0,1)|}{\sqrt{0^2+1^2+1^2} \cdot \sqrt{0^2+0^2+1^2}} = \frac{1}{\sqrt{2} \cdot 1} = \frac{1}{\sqrt{2}} = \frac{\pi}{4} \text{ radians}$$

b) $V: -x = y = \frac{z-1}{2}, \quad \pi: 2x - y = 0$

$$x = -1, y = 1, z = 1 \quad \Rightarrow n = (2, -1, 0)$$

$$\vec{V} = (-1, 1, 2)$$

$$\sin \theta = \frac{|(-1, 1, 2) \cdot (2, -1, 0)|}{\sqrt{(-1)^2+1^2+2^2} \cdot \sqrt{2^2+(-1)^2+0^2}} = \frac{3}{\sqrt{6} \cdot \sqrt{5}} = \frac{3}{\sqrt{30}} = \frac{\sqrt{30}}{10}$$

$$\arcsin \theta = \frac{\sqrt{30}}{10}$$

c) $V: x = (1,0,0) + 1(1,1,-2), \quad \pi: x + y - z - 1 = 0$

$$\vec{V} = (1, 1, -2)$$

$$n\pi = (1, 1, -1)$$

$$\sin \theta = \frac{|(1, 1, -2) \cdot (1, 1, -1)|}{\sqrt{1^2+1^2+(-2)^2} \cdot \sqrt{1^2+1^2+(-1)^2}} = \frac{4}{\sqrt{6} \cdot \sqrt{3}} = \frac{4}{3\sqrt{2}} = \frac{2\sqrt{2}}{3}$$

$$\pi = \arcsin \left(\frac{2\sqrt{2}}{3} \right)$$

4- $\pi_1: x + y + z = 0$

$$n_1 = (1, 1, 1)$$

$$\vec{V} \cdot \vec{n}_1 = 0$$

$$V_x + V_y + V_z = 0$$

$45^\circ: \pi_2: x - y = 0$

$$n_2 = (1, -1, 0)$$

$$\sin 45^\circ = |\vec{V} \cdot \vec{n}_2|$$

$$\frac{|\vec{V}| \cdot |\vec{n}_2|}{\sqrt{2}} = \frac{|V_x - V_y|}{\sqrt{2}}$$

$$|V_x - 2V_y| = 1$$

$$\begin{cases} u_x + u_y + u_z = 0 \\ |u_x - u_y| = 1 \end{cases} \Rightarrow \begin{cases} 2u_y + u_z = -1 \\ u_x = u_y - 1 \end{cases} \Rightarrow \begin{cases} u_x = 1 \\ u_y = 0 \\ u_z = -1 \end{cases}$$

$$s: (1, 0, -1)$$

$$\|\vec{u}\| = \sqrt{1^2 + 0^2 + (-1)^2} = \sqrt{2} \quad \vec{u} = \frac{\vec{s}}{\|\vec{s}\|} = \left(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}} \right)$$

$$5. a) \pi_1: 2x + y - z - 1 = 0 \quad \pi_2: (x - y + 3z - 10 = 0) \\ n_1 = (2, 1, -1) \quad n_2 = (1, -1, 3)$$

$$\cos \theta = \frac{(2, 1, -1) \cdot (1, -1, 3)}{\sqrt{2^2 + 1^2 + (-1)^2} \cdot \sqrt{1^2 + (-1)^2 + 3^2}} = \frac{-2}{\sqrt{6} \cdot \sqrt{11}} = \frac{-2}{\sqrt{66}} \quad \text{arccos} = \frac{2}{\sqrt{66}}$$

$$b) x = (1, 0, 0) + \lambda(1, 0, 1) + \mu(-1, 0, 0) \quad \pi_2: x + y + z = 0 \\ n = \begin{vmatrix} i & j & k \\ 1 & 0 & 1 \\ -1 & 0 & 0 \end{vmatrix} = (0, -1, 0) \quad n = (1, 1, 1)$$

$$\cos \theta = \frac{(0, -1, 0) \cdot (1, 1, 1)}{\sqrt{0^2 + (-1)^2 + 0^2} \cdot \sqrt{1^2 + 1^2 + 1^2}} = \frac{-1}{\sqrt{3}} = \frac{\sqrt{3}}{3} \quad \text{arccos} = \frac{\sqrt{3}}{3}$$

$$c) \pi_1: x = (0, 0, 0) + \lambda(1, 0, 0) + \mu(1, 1, 1) \quad \pi_2: x = (1, 0, 0) + \lambda(-1, 2, 0) + \mu(0, 1, 0) \\ n_1 = \begin{vmatrix} i & j & k \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{vmatrix} = (0, -1, 1) \quad n_2 = \begin{vmatrix} i & j & k \\ -1 & 2 & 0 \\ 0 & 1 & 0 \end{vmatrix} = (0, 0, -1)$$

$$\cos \theta = \frac{(0, -1, 1) \cdot (0, 0, -1)}{\sqrt{0^2 + (-1)^2 + 1^2} \cdot \sqrt{0^2 + 0^2 + 1^2}} = \frac{-1}{\sqrt{2} \cdot 1} = \frac{-1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \quad \text{arccos} = \frac{\pi}{4}$$

$$d - 2x - y + z = 0$$

$$n_1 = (2, -1, 1) \quad n_2 = (1, -2, 1)$$

$$\cos \theta = \frac{(2, -1, 1) \cdot (1, -2, 1)}{\sqrt{2^2 + (-1)^2 + 1^2} \cdot \sqrt{1^2 + (-2)^2 + 1^2}} = \frac{5}{\sqrt{6} \cdot \sqrt{6}} = \frac{5}{6} \quad \text{arccos} = \frac{5}{6}$$

7. a) $r: x-1 = 2y = z$, $A = (1, 1, 0)$, $B = (0, 1, 1)$

$x = 1+t, y = t/2, z = t$

$d(PA) = \sqrt{(1+t-1)^2 + (t/2-1)^2 + (t-0)^2} = \sqrt{t^2 + (t/2-1)^2 + t^2}$

$d(PB) = \sqrt{(1+t-0)^2 + (t/2-1)^2 + (t-1)^2} = \sqrt{(t+1)^2 + (t/2-1)^2 + (t-1)^2}$

$dPA = dPB \Rightarrow \sqrt{t^2 + (t/2-1)^2 + t^2} = \sqrt{(t+1)^2 + (t/2-1)^2 + (t-1)^2}$

$2t^2 + (t/2-1)^2 = (t+1)^2 + (t/2-1)^2 + (t-1)^2$

$2t^2 = t^2 + 2t + 1 + t^2 - 2t + 1$

$2t^2 = 2t^2 + 2$

$0 = 2$ falso

$\therefore$ Não existem pontos equidistantes a A e B na reta $\checkmark$

b) $r(0, 0, 4) + \lambda(4, 2, -3)$, $A = (2, 2, 5)$ e $B = (0, 0, 1)$
 $P = (4\lambda, 2\lambda, 4-3\lambda)$

$d(PA) = \sqrt{(4\lambda-2)^2 + (2\lambda-2)^2 + (4-3\lambda-5)^2}$

$d(PB) = \sqrt{(4\lambda-0)^2 + (2\lambda-0)^2 + (4-3\lambda-1)^2} = \sqrt{(4\lambda)^2 + (2\lambda)^2 + (-3\lambda+3)^2}$

$\sqrt{(4\lambda-2)^2 + (2\lambda-2)^2 + (-3\lambda-1)^2} = \sqrt{(4\lambda)^2 + (2\lambda)^2 + (3-3\lambda)^2}$

$(16\lambda - 16\lambda + 4) + (4\lambda^2 - 8\lambda + 4) + (9\lambda^2 + 6\lambda + 1) = 16\lambda^2 + 4\lambda^2 + (9 - 18\lambda + 9\lambda^2)$

$29\lambda^2 - 18\lambda + 9 = 29\lambda^2 - 18\lambda + 9$

$0 = 0$ Verdadeiro

$\therefore$ todos os pontos da reta r são equidistantes a A e B

c) $r: x = (2, 3, -5) + \lambda(1, 1, 1)$, $A = (1, 1, 0)$, $B = (2, 2, 4)$
 $P = (2+\lambda, 3+\lambda, -5+\lambda)$

$dPA = \sqrt{(2+\lambda-1)^2 + (3+\lambda-1)^2 + (-5+\lambda-0)^2} = \sqrt{(\lambda+1)^2 + (\lambda+2)^2 + (\lambda-5)^2}$

$dPB = \sqrt{(2+\lambda-2)^2 + (3+\lambda-2)^2 + (-5+\lambda-4)^2} = \sqrt{\lambda^2 + (\lambda+1)^2 + (\lambda-9)^2}$

$\sqrt{(\lambda+1)^2 + (\lambda+2)^2 + (\lambda-5)^2} = \sqrt{\lambda^2 + (\lambda+1)^2 + (\lambda-9)^2}$

$3\lambda^2 + 14 = 3\lambda^2 - 12\lambda + 50 \Rightarrow \lambda = 3$

Para $\lambda = 3$

$$P = (2+3, 3+3, -3+3) = (5, 6, 0)$$

é o ponto equidistante de A e B no reta r e $(5, 6, 0)$

8-a) $P = (-2, 0, 1)$, $r: X = (1, -2, 0) + \lambda(3, 2, 1)$

$$A = (1, -2, 0) \quad \vec{v} = (3, 2, 1)$$

$$\vec{AP} = P - A = (-2 - 1, 0 - (-2), 1 - 0) = (-3, 2, 1)$$

$$\vec{AP} \times \vec{v} = \begin{vmatrix} i & j & k \\ -3 & 2 & 1 \\ 3 & 2 & 1 \end{vmatrix} = (0, 6, -12)$$

$$\|\vec{AP} \times \vec{v}\| = \sqrt{0^2 + 6^2 + (-12)^2} = \sqrt{180} = 6\sqrt{5}$$

$$\|\vec{v}\| = \sqrt{3^2 + 2^2 + 1^2} = \sqrt{14}$$

$$D = \frac{6\sqrt{5}}{\sqrt{14}} = \frac{6\sqrt{70}}{14} = \frac{3\sqrt{70}}{7}$$

$$D = \frac{\|\vec{AP} \times \vec{v}\|}{\|\vec{v}\|}$$

5) $P = (1, -1, 4)$, $r: \frac{x-2}{4} = \frac{y}{-3} = \frac{z-1}{2}$

$$x = 2 + 4\lambda, \quad y = -3\lambda, \quad z = 1 + 2\lambda$$

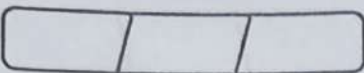
$$A = (2, 0, 1) \quad \vec{v} = (4, -3, 2)$$

$$\vec{AP} = P - A = (1-2, -1-0, 4-1) = (-1, -1, 3)$$

$$\vec{AP} \times \vec{v} = \begin{vmatrix} i & j & k \\ -1 & -1 & 3 \\ 4 & -3 & 2 \end{vmatrix} = (11, 10, 7) \Rightarrow \sqrt{11^2 + 10^2 + 7^2} = \sqrt{270} = 3\sqrt{30}$$

$$\|\vec{v}\| = \sqrt{4^2 + (-3)^2 + 2^2} = \sqrt{29}$$

$$D = \frac{3\sqrt{30}}{\sqrt{29}} = \frac{3\sqrt{870}}{29}$$



$$c) P = (0, -1, 0)$$

$$r: x = 2y - 3 = 2z - 1$$

$$x = 1, y = \frac{1+3}{2}, z = \frac{1+1}{2}$$

$$A = (0, 3/2, 1/2) \quad \vec{v} = (1, 1/2, 1/2)$$

$$\vec{AP} = A - P = (0 - 0, -1 - 3/2, 0 - 1/2) = (0, -5/2, -1/2)$$

$$\vec{AP} \times \vec{v} = \begin{vmatrix} i & j & k \\ 0 & -5/2 & -1/2 \\ 1 & 1/2 & 1/2 \end{vmatrix} = (-1, -1/2, 5/2)$$

$$\|\vec{AP} \times \vec{v}\| = \sqrt{(-1)^2 + (-1/2)^2 + (5/2)^2} = \sqrt{30}/2$$

$$\|\vec{v}\| = \sqrt{1^2 + (1/2)^2 + (1/2)^2} = \sqrt{6}/2$$

$$D = \frac{\sqrt{30}/2}{\sqrt{6}/2} = \frac{\sqrt{30}}{\sqrt{6}} = \sqrt{5}$$

$$9.) \begin{cases} \pi_1: x + y = 2 \\ \pi_2: x = y + z \end{cases} \Rightarrow \begin{cases} x = 2 - y \\ z = 2 - 2y \end{cases} \Rightarrow (2 - y, y, 2 - 2y)$$

$$S = (x, y, z) \Rightarrow \vec{v} = (1, 1, 1)$$

$$\vec{AP} = (2 - 1, 1, 3 - 2 \cdot 1)$$

$$\vec{AP} \times \vec{v} = \begin{vmatrix} i & j & k \\ 2-1 & 1 & 3-2 \\ 1 & 1 & 1 \end{vmatrix} = (3 \cdot 1 - 3, 1 - 1, 2 - 2 \cdot 1)$$

$$\|\vec{AP} \times \vec{v}\| = \sqrt{(3 \cdot 1 - 3)^2 + (1 - 1)^2 + (2 - 2 \cdot 1)^2} = \sqrt{14 \cdot 1^2 - 28 \cdot 1 + 14} = \sqrt{14} |1 - 1|$$

$$\|\vec{v}\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$D = \frac{\sqrt{14} |1 - 1|}{\sqrt{3}} = \sqrt{\frac{14}{3} |1 - 1|} = \frac{14}{3} \Rightarrow \|A - 1\| = 1$$
$$1 = 2 \text{ or } 1 = 0$$

para $x=2 \Rightarrow DP = (2-2, 2, 2-4) = (0, 2, -2)$

para $x=0 \Rightarrow DP = (2-0, 0, 2-0) = (2, 0, 2)$

As pontos são $(0, 2, -2)$ e $(2, 0, 2)$

10-a) $P = (1, 3, 4)$ $\pi: X = (1, 0, 0) + \lambda(1, 0, 0) + \mu(-1, 0, 3)$

$$n = \begin{vmatrix} i & j & k \\ 1 & 0 & 0 \\ -1 & 0 & 3 \end{vmatrix} = (0, 3, 0)$$

$0(x-1) + 3(y-0) + 0(z-0) = 0$

$3y = 0$

$y = 0$

$$D = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

Para $y=0$

$$D = \frac{|0 \cdot 1 + 1 \cdot 3 + 0 \cdot 4 + 0|}{\sqrt{0^2 + 1^2 + 0^2}} = \frac{3}{1} = 3$$

b) $P = (0, 0, -6)$ $\pi: X - 2y - 2z - 6 = 0$

$$D = \frac{|1 \cdot 0 - 2 \cdot 0 - 2 \cdot (-6) - 6|}{\sqrt{1^2 + (-2)^2 + (-2)^2}} = \frac{6}{\sqrt{9}} = \frac{6}{3} = 2$$

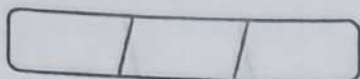
c) $P = (1, 1, 1)$ $\pi: 2x - y + 2z - 3 = 0$

$$D = \frac{|2 \cdot 1 - 1 \cdot 1 + 2 \cdot 1 - 3|}{\sqrt{2^2 + (-1)^2 + 2^2}} = \frac{0}{\sqrt{9}} = \frac{0}{3} = 0$$

$$\begin{cases} 11- r = x - 2y = y + z \\ x = 2 - y \Rightarrow z = 2 - 2y \\ x = y + z \end{cases} \Rightarrow \begin{cases} x = 2 - 1 \\ y = 1 \\ z = 2 - 2 \cdot 1 \end{cases}$$

$\pi: X - 2y - z = 1$

$$D = \frac{|x_0 - 2y_0 - z_0 - 1|}{\sqrt{1^2 + (-2)^2 + (-1)^2}} = \frac{|x_0 - 2y_0 - z_0 - 1|}{\sqrt{6}}$$



$$D = \sqrt{6}, \text{ então}$$

$$\frac{|x_0 + 2y_0 - 2z_0 - 1|}{\sqrt{6}} = \sqrt{6} \Rightarrow |x_0 + 2y_0 - 2z_0 - 1| = 6$$

$$|(2 - \lambda - 2\lambda - (2 - 2\lambda) - 1)| = 6$$

$$-1 - \lambda = 6$$

$$\lambda = -7 \text{ ou } \lambda = 5$$

$$\text{Para } \lambda = -7 \Rightarrow P_1 = (2 - (-7), (-7), 2 - 2(-7)) = (9, -7, 16)$$

$$\text{Para } \lambda = 5 \Rightarrow P_2 = (2 - 5, 5, 2 - 2 \cdot 5) = (-3, 5, -8)$$

$$\text{Os pontos são } (9, -7, 16) \text{ e } (-3, 5, -8)$$

$$12 - a) \quad r = x(2, 1, 0) + \lambda(1 - 1, 1) \quad , \quad s: x + y + z = 2 \Rightarrow 5 \cdot (0, -1, 3) + \lambda(1, 2, 3)$$

$$V_r = (1, -1, 1) \quad \text{Não paralelos}$$

$$V_s = (1, 2, 3)$$

$$\begin{cases} x + y + z = 2 \\ 2x - y = 1 \end{cases} \Rightarrow \begin{matrix} 5 \cdot (0, -1, 3) + \lambda(1, 2, 3) \\ 2x - y = 1 \end{matrix}$$

$$D(r, s) = \frac{|(P_s - P_r) \cdot (V_r \times V_s)|}{\|V_r \times V_s\|} \quad \Rightarrow \quad (2, 1, 0) - (0, -1, 3) = (2, 2, -3)$$

$$\|V_r \times V_s\|$$

$$\frac{|1|}{\sqrt{26}} = \frac{\sqrt{26}}{26}$$

$$\begin{aligned} & (-2)(1) + (-2)(4) + (-3)(3) = -1 \\ & \sqrt{(-2)^2 + (-2)^2 + (-3)^2} = \sqrt{17} \end{aligned}$$

$$b) \quad r = \frac{x+4}{3} = \frac{y}{4} = \frac{z+5}{-2} \quad , \quad s: x = (2, -5, 2) + \lambda(6, 4, -1)$$

$$V_r = (3, 4, -2) \quad \text{Não paralelos}$$

$$V_s = (6, 4, -1)$$

$$\begin{vmatrix} i & j & k \\ 3 & 4 & -2 \\ 6 & 4 & -1 \end{vmatrix} = (4, 7, 12)$$

$$D = \frac{169}{36} = 13$$

$$c) r: \frac{x-1}{-2} = 2y = z \quad s: x = (0, 0, 2) + \lambda(-4, 1, 2) + \mu(1, 0, -1)$$

$$x = -2\lambda + 1, y = \lambda/2, z = \lambda$$

$$v_r = (-4, 1, 2) \quad v_s = (-4, 1, 2) \quad \left. \begin{array}{l} \\ \end{array} \right\} \text{paralela}$$

$$\frac{\| (P_s - P_r) \times v \|}{\| v \|} = \frac{\| (0, 0, 2) - (1, 0, 0) \| \cdot \| (-4, 1, 2) \|}{\| (-4, 1, 2) \|} = \frac{\sqrt{96}}{21}$$

$$13-a) r: x = (1, 9, 4) + \lambda(3, 3, 3) \quad \pi: x = (5, 7, 9) + \lambda(1, 0, 0) + \mu(0, 0, 1)$$

$$v \cdot n = (3, 3, 3) \cdot (0, 0, 1) = 3 \quad \begin{array}{c|ccc|c} n & i & j & k & \\ \hline \text{Não} & 1 & 0 & 0 & -(0, 0, 1) \\ \text{paralela} & 0 & 1 & 0 & \end{array}$$

Distância 0, paralela

$$7) r: x - y + z = 0 \quad 2x + y - z = 3 \quad y - z = 10$$

$$\left. \begin{array}{l} x - y + z = 0 \\ 2x + y - z = 3 \end{array} \right\} \Rightarrow (1, y, y-1) + (1, 0, 1) = (2, y+1, y) \quad n = (0, 1, -1)$$

$$v \cdot n = (0, 1, 1) \cdot (0, 1, -1) = 0 \quad \text{paralela}$$

$$D = \frac{|0 \cdot 1 + 1 \cdot 0 + (-1) \cdot (-1) - 4|}{\sqrt{0^2 + 1^2 + (-1)^2}} = \frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

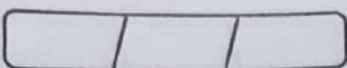
$$c) r: x = y - 1 = z + 3 \quad \pi: 2x + y - 3z - 12 = 0$$

$$r: (0, 1, -3) + t(1, 1, 1) \quad n = (2, 1, -3)$$

$$v \cdot n = (1, 1, 1) \cdot (2, 1, -3) = 0 \quad \text{paralela}$$

$$2 \cdot 0 + 1 - 3(-3) - 12 = 0 \quad 0 + 1 + 9 - 12 = 0 \quad (\text{controle no plano})$$

$$D = 0$$



$$14-a) \pi_1: 2x - y + 2z + 0 = 0$$

$$n_1 = (2, -1, 2)$$

$$\pi_2: 4x - 2y + 4z - 21 = 0$$

$$n_2 = (4, -2, 4)$$

$$n_1 = \lambda n_2 \Rightarrow \text{paralelos}$$

$$\lambda = 0,5$$

$$D = |D_1 - 0| \Rightarrow \frac{|-10,5 - 0|}{\sqrt{2^2 + (-1)^2 + 2^2}} = \frac{10,5}{3} = \frac{7}{2}$$

$$b) \pi_1: 2x + 2y + 2z = 5$$

$$n_1 = (2, 2, 2)$$

não múltiplos

$$n_1 \text{ e } n_2 \text{ não paralelos}$$

$$D = 0$$

$$\pi_2: X = (2, 2) + 1(-1, 0, 3) + \mu(1, 1, 0)$$

$$n = \begin{vmatrix} i & j & k \\ -1 & 0 & 3 \\ 1 & 1 & 0 \end{vmatrix} = (-3, 3, -1)$$

$$c) \pi_1: x + y + z = 0$$

$$n_1 = (1, 1, 1)$$

$$n_1 \text{ e } n_2 \text{ não paralelos}$$

Não múltiplos

$$D = 0$$

$$\pi_2: 2x + y + z + 200 = 0$$

$$n_2 = (2, 1, 1)$$

15- π

$$v: x + z = 5 = y + 4$$

$$y = -4, x = t, z = 5 - t$$

$$v_r = (1, 0, -1)$$

$$s: (4, 1, 1) + \lambda(4, 2, -3)$$

$$v_s = (4, 2, -3)$$

$$(t, -4, 5-t) = (4+4\lambda, 1+2\lambda, 1-3\lambda)$$

$$\begin{cases} t = 4+4\lambda \\ -4 = 1+2\lambda \Rightarrow \lambda = -5/2 \\ 5-t = 1-3\lambda \end{cases}$$

$$t = -6$$

$$\Rightarrow 5 - (-6) = 1 - 3(-5/2)$$

$$\pi: 2x - y + 2z - 3 = 0 \text{ e } 2x - y + 2z - 15 = 0$$